The Project, Complete

Findings, sealed evidence, open questions and the asks — 22 August 2026

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This is the whole project in a few pages, written so it can be read instead of the paper. The paper (10 pages, IEEE format) exists and is current; everything here points at a result that is committed in the repository, and every number below is re-derived from stored results by an automated check before it is allowed to appear in a document.

The question, and why it matters

A fairness constraint asks a model to give two groups similar approval rates, but it does not say how the gap should close. It can close because more people from the disadvantaged group are approved, or because people from the advantaged group lose approvals they would have had. Both count as success, and the fairness metric — the number a company would report and a regulator would check — comes out the same in both cases. This means a team imposing a constraint cannot tell, from its own dashboard, whether it is about to extend opportunities or withdraw them.

The literature already knew the metric was silent about this. What it did not have is a way to know **in advance** which of the two outcomes a given system will produce. That is what this project measured, and the answer turns out to be readable from one number every deployed system already has: the rate at which its current model says yes.

The finding

Below a crossover in the baseline approval rate — empirically near 0.54 — an attribute-blind parity constraint shrinks the total pool of favourable decisions, while above it the same constraint grows the pool. On UCI Adult (rate 0.24), six different mitigation methods all shrink the pool, by 7.9% to 22.1%. On mortgage approvals (rate 0.81), the identical constraint grows it by 4.3%. The parity metric reports success identically in both cases, which is exactly the problem: the direction is invisible to the certificate but predictable from the rate.

The claim is deliberately narrow, and each boundary is measured rather than assumed. It holds within a population, in the attribute-blind in-processing regime, and it predicts the direction only — magnitude is ordered within a population (Spearman rho up to +0.96) but does not transfer across populations, where a pooled slope reaches only +0.487 against a pre-registered bar of +0.70 and therefore fails.

Why the evidence is stronger than a correlation

Every correlation in the project was computed after the arms were run, which is the weakest support for a claim of the form "you can know in advance". So the rule was tested the only way that format can be tested: **sealed predictions, committed to git before the data existed.**

The first sealed test failed, and the failure was informative. A refinement added from four in-sample populations two hours before the seal turned what would have been 7 of 8 into **4 of 8**, no better than a constant. The refinement was withdrawn, and the simple rule — down below 0.54, up at or above — was re-sealed against ten states the project had never touched, with the bar (at least 9 of 10, and strictly beating the best constant) committed before any run.

It scored 9 of 10, against a best constant of 6. The one miss is charged against the rule under the criterion sealed with it, and it is an instructive miss: an arm whose true effect is around −0.04% and flips sign between random seeds, which means there may have been no direction there to predict.

The complete ledger of registered tests is below, and one thing matters for reading it: these are not seven attempts at one claim, of which two succeeded. Each test targeted a **different** claim, and each failure removed a claim *stronger* than the one that survives — so the failures are boundary measurements, and the passes test the claim as those boundaries define it. Averaging the rows into "2 out of 7" would be reading a map as a scorecard.

Test	Outcome	What it settled
HMDA held-out sweep	2 of 4 — fail	our measuring procedure is unreliable on mortgage data, so no claim rests on it there
Equalized-odds transfer	+0.334 vs +0.70 — fail	the rule works for parity, which controls approval rates directly; it failed for equalized odds, which does not
Post-processing transfer	r = −0.024 — fail	the rule stops working the moment the model may read the protected attribute — and that turned out to be the boundary the theory predicts
Selection-rate derivation	beaten by a constant — fail	we cannot explain **why** the rule works; the paper claims only that it does
Sealed rule, refined	4 of 8 — fail	the extra clause we had added from early data was wrong, and is withdrawn
Attribute-aware replication	9 of 9 — **holds**	where the attribute is readable, the disadvantaged group always gains — as the theory says, predicted in advance
Re-seal, unrefined rule	9 of 10, constant 6 — **holds**	the simple rule predicts populations it has never seen, from the approval rate alone

Five failures are reported uncorrected, which removes the usual way lucky results reach print: selective reporting. And the passes are unlikely to be luck on their own terms — 9 of 10 against a guesser as good as the best constant happens about one time in twenty by chance, the arms were deliberately spread so no constant strategy could score well, and the refined rule, which looked equally convincing in-sample, was given the same sealed test and failed it.

The relationship to the 2026 theory paper

Fairness May Backfire (arXiv:2603.06901) proves that in the attribute-blind regime the direction is distribution-dependent, so the conditionality itself is theirs and the paper says so. What they do not have is any way to compute the direction: their conditions are stated over a quantity that diverges on real data, and on 26 real arms from 15 populations they can be evaluated on none. Their paper contains no experiments and no occurrence of "selection rate" or "base rate", which was checked against the full text.

The project's position relative to them is two-sided, and both sides are measured. Their regime distinction is right: under attribute-aware post-processing the selection-rate relationship vanishes entirely (r = −0.024 against +0.585 on the same populations), and their prediction for that regime — the disadvantaged group always gains — holds on **27 of 27 populations, nine of them pre-registered**. And their structural quantity is tracked by the plain selection rate at r = +0.935, which means the number this project uses is a computable proxy for the quantity their theorem is stated over. They proved the question has an answer; this project built the instrument that reads it.

Who actually pays, which the metric also hides

Measured in rates, the burden of a fairness fix looks nearly even. Measured in people it is not: on Adult, between 66% and 74% of the individuals whose decision changed were advantaged-group members losing an approval, at 2.68 approvals destroyed per one created. Adding a one-line floor to the objective — "do not shrink the pool" — drops that exchange rate to 0.88 across nineteen arms at a cost of about 0.12 accuracy points, so the harm is optional once you know to ask for it. Knowing when to ask is what the crossover rule is for.

Scale, and the discipline behind the numbers

The record holds **47 independent populations** across seven data sources (UCI Adult, ACS states, HMDA mortgages, COMPAS, LSAC, the Dutch census, Taiwanese credit-card default), six decision domains, three countries and two decades. The last ten arrived through a pre-registered sweep whose verdicts landed the same day this report was finished, and both went against us in instructive ways: the test of what predicts the crossover's location returned **underpowered** (three located against a floor of six), a sealed model of the effect's *size* lost outright to predicting zero, and the three crossovers that did locate **broke the cluster** — Florida at 0.284 and Pennsylvania at 0.652, far outside the 0.43–0.58 band, on the same instrument that built it. Several of the largest states turn out to have no crossover at all, and the fixed 0.54 prior scores 5 of 10 on their natural arms, each miss agreeing with that state's own measured crossover. So the within-population claim survived the week's hardest test, while the transferable prior narrowed to "usually, not always" — and every one of those sentences comes from a prediction committed before the data existed.

Three pieces of discipline are worth naming because they caught real errors:

- **55 automated checks** re-derive every documented figure from stored results; one of them caught the paper's population count going stale within minutes of new results landing today.
- **Population counts are recomputed, never quoted** — three counts in the project's own history turned out to be arm counts masquerading as population counts, and the paper now carries a table saying exactly which populations enter which analysis.
- **Anything that could go either way is sealed first.** The standing rule commits predictions, thresholds and the naive baseline to beat before the run; re-analysis of existing results is labelled post-hoc every time it appears.

What is open, honestly

- **Why the crossover sits where it sits.** Located values now span 0.28–0.65 and nothing predicts the location; the sealed test built to answer this returned underpowered, partly because a third of the large states have no crossover to locate — their response to the constraint is U-shaped or positive throughout, which is itself the newest open question.

- **Breadth.** The sealed 9-of-10 is ten US census populations — depth, not breadth. A non-Western census extract (Brazil 2000/2010, Mexico 2015/2020) is requested and pending approval, which would allow the first crossover located outside the West.
- **Mechanism.** At extreme operating points the constraint closes the gap by two qualitatively different solutions — lifting the disadvantaged versus discarding approvals — and what selects between them is measured to be none of six candidate quantities. The paper states this as an open boundary rather than papering over it.

What I am asking

1. **Venue and page limit.** The paper is 10 pages in IEEE conference format; FAccT or AIES would want roughly 10, IEEE conferences 8. This decides what gets cut.
2. **Does the empirical half stand as a paper, given the theory paper exists?** My position: their theorem is uncomputable on data and this project supplies the measurement, the boundary verification and a sealed predictive rule — but this is the call I need from you.
3. **AI disclosure.** How the department wants assistance disclosed, including whether commits should carry a trailer; none currently do.
4. **Whether the course submission window is still open** for the rebuilt report and deck.
5. **Contacting the theory paper's authors.** Our results confirm their regime prediction (27 of 27, nine pre-registered), extend their framework with a computable proxy for its central quantity, and show its stated conditions cannot be evaluated on real data — the last of which rests on an estimator choice they would be the best people to challenge before a reviewer does. Standard practice would be a short collegial email, ideally after a preprint exists; I would like your read on whether and when, and whether you want to be on it.